Notation.

Let $X: U \to \mathbb{R}^3$ be a Patch. Define

$$g_{ij} \overset{\text{def}}{=} \langle X_i, X_j \rangle$$

So that given two tangent vectors $A = \sum a^i X_i$ and $B = \sum b^j X_j$ satisfies

$$\langle A, B \rangle = \sum_{1 \le i,j \le 2} a^i b^j \cdot \langle X_i, X_j \rangle = \begin{pmatrix} b^1 & b^2 \end{pmatrix} \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \cdot \begin{pmatrix} a^1 \\ a^2 \end{pmatrix}$$

And, $g \overset{\text{def}}{=} \det(g_{ij})$ and $g^{kl} = (k,l)$ entry of $(g_{ij})^{-1}$.

The normal of X at $X(s)$ is $(s = (a,b))$

$$n(s) = \frac{X_1(s) \times X_2(s)}{\| X_1(s) \times X_2(s) \|}.$$

And, the second-derivatives at $X(s)$ is

$$X_{ij}(s) = \frac{\partial^2 X}{\partial u^j \partial u^i}(s)$$

And, the instrinsic normal of a unit speed curve in the image of X is

$$S = n \times T.$$

The Normal and Geodesic Curvature of a Unit Speed Curve γ in X

$$k_n = \langle \gamma''(s), n(\gamma^1(s), \gamma^2(s)) \rangle \quad \text{and} \quad k_g = \langle \gamma''(s), S(s) \rangle$$

The coefficients of the second fundamental form of X is

$$L_{ij} = \langle X_{ij}, n \rangle \qquad (1 \le i, j \le 2)$$

The Christoffel symbols

$$\Gamma^k_{ij} \overset{\text{def}}{=} \sum_{l=1}^{2} \langle X_{ij}, X_l \rangle g^{lk}.$$